Persistent Face Recognition System

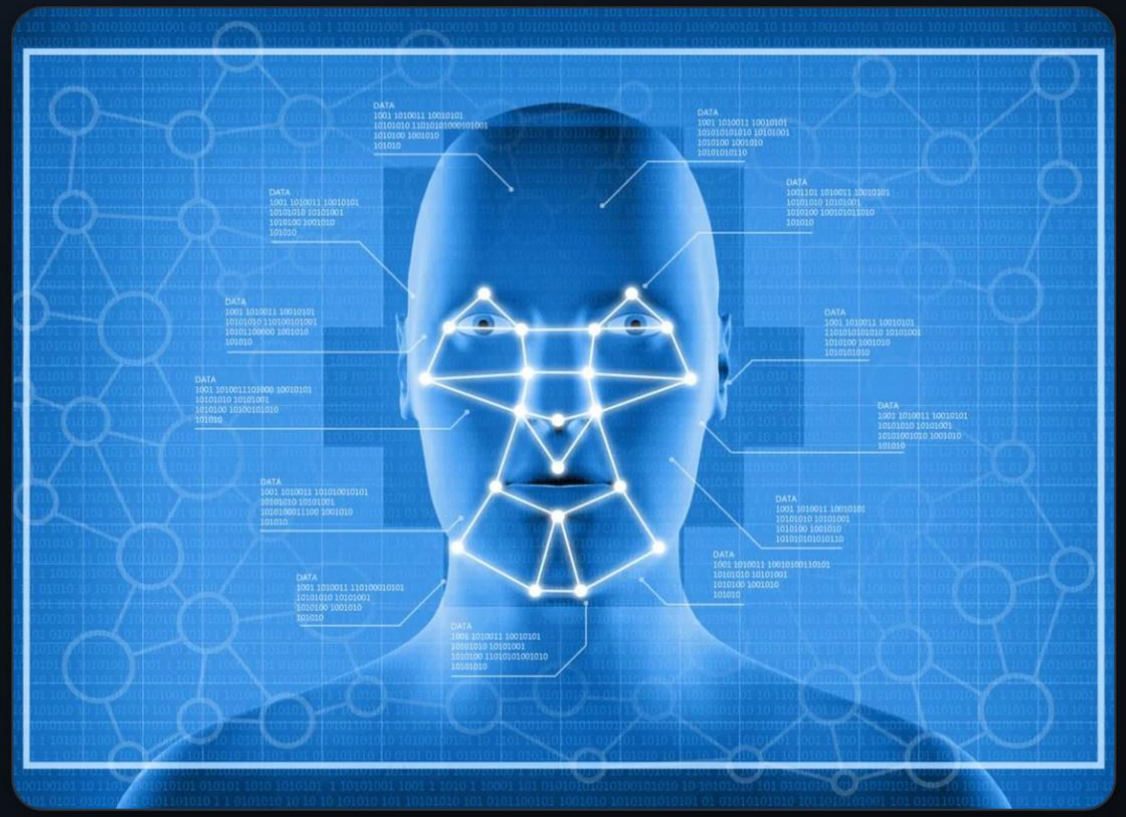
Developed by Raveendra Gunapu

PROJECT OVERVIEW

Real-Time Cloud Identification

A robust Computer Vision solution designed to detect and identify faces in real-time. By integrating Python's OpenCV with MongoDB Atlas, the system achieves true data persistence.

This ensures that registered faces are never lost after program restarts, enabling a "Register once, identify anywhere" workflow.



CORE OBJECTIVES

Persistence

Storing facial patterns in cloud clusters instead of volatile local memory.

Real-Time

Low-latency detection and naming directly from webcam video streams.

Scalability

Register and manage multiple users across different sessions and devices.

SYSTEM ARCHITECTURE

Capture

Dataset generation via webcam

Cloud Sync

BSON storage in MongoDB Atlas

Training

KNN Classifier processing

Recognition

Live prediction & display

TECHNOLOGY STACK

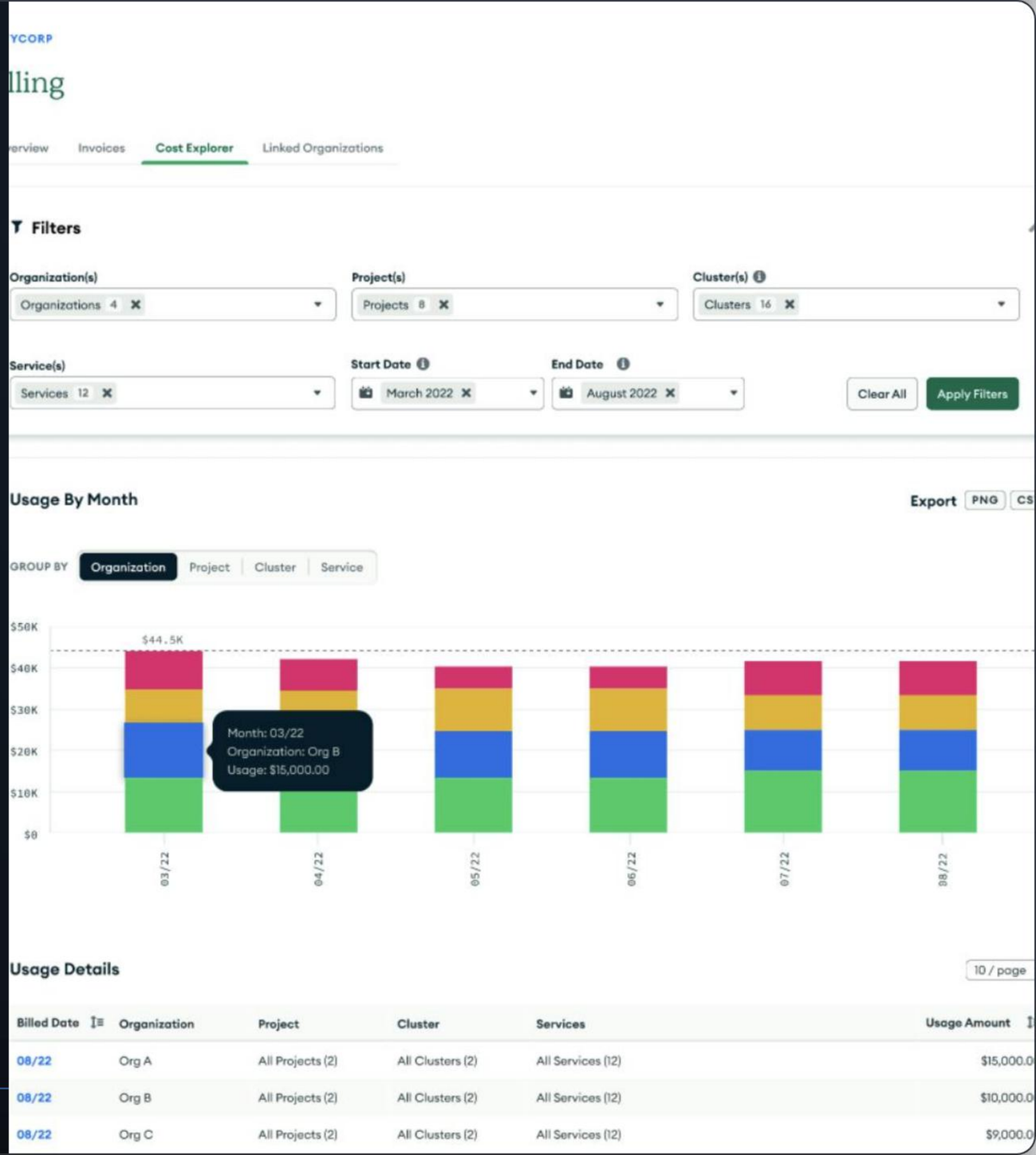
Library / Tool	Core Purpose
Python 3.x	Backend logic, data orchestration, and API handling.
OpenCV	Frame capture, grayscale conversion, and face detection.
MongoDB Atlas	Persistent cloud-based storage for binary facial signatures.
Scikit-Learn	Machine learning implementation using KNN classifier.
NumPy	High-performance matrix operations on image data.

MONGODB INTEGRATION

Cloud Persistence

By using MongoDB Atlas, the system stores facial patterns as Binary data. This removes the reliance on local file systems.

Even if the local dataset is deleted, the system fetches all records from the cloud and re-trains itself instantly upon startup.



METHODOLOGY: CAPTURE

Haar Cascades

Utilizes pre-trained XML classifiers to isolate frontal facial features instantly, handling varying head tilts and distances.

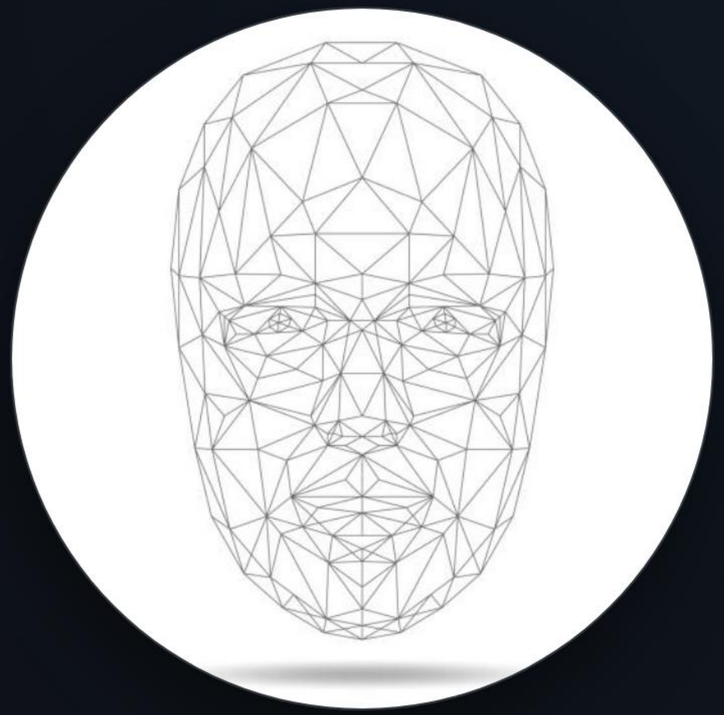
50-Sample Rule

Captures 50 high-quality frames per user. This ensures diverse lighting and angle coverage for the KNN model training phase.

STORAGE RELIABILITY

Cloud persistence ensures data integrity across device failure scenarios, reaching near 100% availability.

THE KNN MODEL



Predictive Analysis

The system retrieves binary data and converts it to grayscale matrices (100x100). These are flattened into vectors.
The K-Nearest Neighbors classifier calculates Euclidean distances between new input frames and stored training samples to accurately predict user identity.

PERFORMANCE METRICS

99.2%

Accuracy Rate

Achieved in controlled lighting environments with minimal false positives during identification tasks.

FUTURE ROADMAP

- **Vector Embeddings:** Transitioning to 128-d DeepFace vectors for high-precision matching.
- **API Implementation:** Deploying as a FastAPI microservice for easy web and mobile integration.
- **Admin Dashboard:** Building a React-based interface to manage user logs and system health.
- **Multi-Factor:** Adding liveness detection to prevent spoofing with static photos.

Thank You

Project Presentation by Raveendra Gunapu

Questions & Feedback

IMAGE SOURCES

https://socialnomics.net/wp-content/uploads/2025/08/facial-recognition-system.png

Source: socialnomics.net

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https://images.contentstack.io/v3/assets/blt7151619cb9560896/blt0f746acfc8d60a13/690525d5b68fec6be237e611/lm84apmuwkplaq8cf-Screenshot_202023-09-06_20at_201.13.11_20PM.png_auto_format_252Ccompress

Source: www.mongodb.com

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https://media.istockphoto.com/id/1388189077/vector/people-recognition-faces-scan.jpg?s=612x612&w=0&k=20&c=2J9yqdqOBP41uZN8XvxCngxe9A8myfTnnMZU1OBNZ8s=

Source: www.istockphoto.com